

The Research Workspace

What every part of it is for, and how to use it.
Written for enrolled researchers on the KuwaitSat-1 programme.

WORKSPACE	/researcher.html, reached by signing in on the mission hub
AUDIENCE	Researchers enrolled on the programme. One account, one workspace.
DATA	KuwaitSat-1 payload frames, held in the mission database and released only inside a signed-in session
VERSION	23 September 2026
KEEP THIS	This document describes the inside of a restricted workspace. It is handed to enrolled researchers. It is not published with the site.

1 What this workspace is, in one page

The KuwaitSat-1 research workspace is where a researcher works with the imagery the satellite actually returned. It does four things, and it is worth being clear about them before any button is pressed.

It releases the payload archive to you. The frames KuwaitSat-1 decoded are held in the mission database. They are not in the public code repository, they are not part of the deployed website, and there is no web address that serves them without a session. When you sign in, the database releases them to your browser. When you sign out, that stops.

It lets you define a mission. A mission is a research question plus the ground it applies to. Both are recorded. The ground is a polygon over Kuwait, drawn on a map, and the database will refuse a polygon that is not over Kuwait.

It runs an orchestrated pipeline against your mission, and it stops halfway for you. Six agents take turns. Three of them run, then everything halts and waits for a human decision. Nothing downstream of that point happens until you press Approve, and your account is recorded as the approver.

It keeps the receipt for all of it. Every step every agent took, every tool it called, every request the database refused, and every finding it produced is a row you can read back. That record is the product. The pictures are evidence; the trail is what makes them usable in research.

The one rule the workspace holds to. Nothing on the screen is invented. If a table is empty, it is because the database returned nothing, and the panel says so in words. If a number is missing, the workspace says it cannot be measured rather than estimating it. If an agent was refused, the refusal is shown, not hidden. A platform whose argument is provenance cannot have a single placeholder row in it.

WHO CAN SEE YOUR WORK

Only you. Missions, runs, agent steps, findings and reports are scoped to your account by a database policy on your user id. Another researcher signed into the same workspace sees their own work and nothing of yours. This is not a filter the page applies, so it cannot be turned off from a browser: it is enforced inside the database, on every request, before any row is sent. The payload archive is the one shared table, and it is read-only for everyone.

2 Getting in

- 1 Go to the mission hub and press **Researcher sign in** in the top right.
- 2 The waiting screen appears while the session is established. Read the notice on it: it states who the workspace is for and that your work is private to your account.
- 3 Enter your email address and password. If you do not have an account, the programme lead creates one for you. There is no self sign-up.
- 4 You land in the research workspace. Signing in always brings you here, not to the public site.

- 5 To go back to the public mission hub, use **Mission hub** in the top right. To end the session everywhere, use **Sign out**, which signs you out of every tab and device.

If you see "This workspace is restricted". Your session was not recognised. Reload the page once: the workspace waits and retries before refusing, but a session arriving from another page can occasionally lose the race. If it happens twice, sign in again from the mission hub.

THE STATUS STRIP

The thin bar under the top navigation is live, not decoration. **PAYLOAD** says whether the archive answered your account. **DATA PIPELINE** says whether the database is reachable at all. If either reads `REFUSED`, stop and tell the team: it means your account reached the database and the database declined, which is different from the site being down.

3 The nine places in the sidebar

The sidebar is grouped the way the work is: operations, then data, then analysis, then outputs. Every item opens a real view and every view reads live rows. The same views are on the top navigation bar, which is what you use on a narrow screen.

3.1 Research Overview

WHAT IT IS FOR

The one screen that answers "where is everything". Open it first and last.

WHAT IS ON IT

PANEL	WHAT IT TELLS YOU
Active Missions	How many missions exist on your account. Nobody else's are counted.
Payload Frames	How many frames the archive released to this session.
Data Ingest	Whether the archive answered. "Refused" is a real answer and means something.
Agent Runs	How many pipeline runs have been started on your missions, ever.
Workflow	The six stages of your most recent run. A filled marker is a step that completed; an outlined one has not been reached.
Map	Every geolocated frame footprint in green, and your own mission areas in white outline.
Agent Operations	The six agents and what each one actually did on the most recent run. Before your first run they all read CONFIGURED, which is honest: they exist and have not been asked to do anything.
Your Missions	The four most recent. Open Missions for the rest.
Session	What this browser has done since you signed in. It is a local log and is not stored anywhere.

HOW TO USE IT

Check the four numbers across the top. If Payload Frames is zero and Data Ingest says Refused, nothing below it will work and the problem is access, not analysis.

3.2 Missions

WHAT IT IS FOR

The list of everything you have asked the platform to investigate, and the place where a single mission is opened, run, decided and closed.

HOW TO USE IT

- 1 Type in the search box to narrow the list. It matches the name, the objective, the status and the area.
- 2 **Click any row** to open the mission. The row highlights and a detail panel opens underneath it.
- 3 The detail panel shows the mission id, its status, the full objective, the area with its size in square kilometres, how many archive frames fall inside that area, a map of the area with those frames drawn on it, and the complete run history.
- 4 Under **Runs and steps** is every step every agent took on this mission, read back out of the database. Refused steps are shown in red with the reason.
- 5 Under **Findings** is everything the run produced.
- 6 The buttons change with the state of the mission. See the table below.

BUTTON	WHEN IT APPEARS, AND WHAT IT DOES
Run the pipeline on this mission	On a mission that is <code>DRAFT</code> or <code>FAILED</code> . Opens the Research Console with this mission selected and starts the run.
Open the checkpoint in the console	On a mission that is <code>QUEUED</code> or <code>RUNNING</code> , which means a run is open and waiting for your decision.
Generate the report	On a mission in <code>REVIEW</code> , which means the pipeline finished and a draft report is waiting. This is the act that signs it.
Show the area on the map	Always. Jumps to Geospatial Layers with the map fitted to this mission's area.
Download .md	Once a report exists. Saves the approved report to your computer as a Markdown file.

WHAT THE STATUSES MEAN

STATUS	MEANING
DRAFT	Created, never run.
QUEUED / RUNNING	A run is open. Either the pipeline is working or it is waiting at the human checkpoint.
REVIEW	The pipeline finished and produced a draft report. It is not a report until you approve it.
COMPLETE	A report has been approved and signed against your account. The mission is closed: it cannot be run again, because evidence behind a signed conclusion must not be able to change after it is signed.
FAILED	The run was abandoned, or the objective was screened as an instruction rather than a question. See section 6.
FLAGGED	Not a status but a marker beside one. The objective on this mission matched the injection screen. See section 6.

3.3 New Mission

WHAT IT IS FOR

Creating the thing everything else hangs off. A mission is a question plus a place.

HOW TO USE IT

- 1 **Mission name.** Between 3 and 120 characters and it must contain letters. Name it after the ground and the question, for example "Kuwait Bay vegetation assessment".
- 2 **Mission type** and **Time window** are optional. They are recorded with the objective.
- 3 **Research objective.** At least 20 characters, at most 1500. Write a *question*. The Mission Orchestrator reads this text and screens it before it calls any tool, and an objective phrased as an order to the system is refused. See section 6.
- 4 **Area of interest.** The map is the field. Whatever is on screen is the area. Pan and zoom until the map frames the ground you mean, and the read-out under it updates live with the corners and the size in square kilometres.
- 5 The six grey buttons above the map jump to named areas: the whole country, Kuwait Bay and the capital, Bubiyan and Khor as-Sabiya, Al-Jahra and the north west, Al-Ahmadi and the south coast, Al-Khiran and Wafra. Use one as a starting point and adjust.
- 6 Press **Create mission**. You land on the mission you just made.

The area is always valid, by construction. The database only accepts a polygon inside 46.5 to 48.8 degrees east and 28.5 to 30.1 degrees north, which is Kuwait and its waters. The map clips your view to that box before it sends it, and tells you when it has. You cannot accidentally create a mission over Saudi Arabia.

Draw the area over ground the satellite actually photographed. An area with no archive frame inside it is a legitimate mission and the pipeline will run, but it will stop at the first step and tell you there is nothing to measure. Open Geospatial Layers first and look at where the frames are.

3.4 Payload Imagery

WHAT IT IS FOR

The archive itself: the frames KuwaitSat-1 returned, each with its acquisition record.

HOW TO USE IT

- 1 The grid shows every frame released to your session, with its number, the place it was matched to, the date it was acquired, its coordinate and a confidence badge.
- 2 **Click a frame** to open the full record sheet underneath: the picture at size, the acquisition date, the geolocation and how it was arrived at, the confidence and the note explaining it, the ground sample distance, the footprint size, the bands, the processing level, the pixel dimensions and the source.

- 3 **Show on the map** takes you to Geospatial Layers with the map fitted to that frame's footprint. It is disabled for a frame with no geolocation.
- 4 **Download the record as CSV** saves the acquisition metadata for every frame. The pictures are deliberately not in that file: the record is yours to take away, the imagery stays inside the session.

THE CONFIDENCE BADGES, AND WHY SOME FRAMES HAVE NO COORDINATE

BADGE	WHAT IT MEANS
high	The coastline in the frame matches the reference imagery closely and unambiguously.
medium	A clear match on distinctive features, with some uncertainty in the exact centre.
low	A plausible match. Treat the coordinate as indicative.
unresolved	No coordinate. The frame was searched for and not found.

The payload record sheet the team was given left the geolocation column empty on every row. Nothing in that column came off the spacecraft. Every coordinate in this archive was produced by matching the coastline in the frame against reference imagery by eye, and each row says so and states how far it should be trusted. Three frames were searched for across the whole of Kuwait at every orientation and scale and did not match anything in the country. They are left empty rather than guessed. That is the right answer, and it is why those frames cannot be used in an analysis.

3.5 Data Archive

WHAT IT IS FOR

A statement of every dataset this workspace can reach, counted live.

HOW TO USE IT

The Records column is a real count from the database at the moment you opened the view. A dataset that says `refused` is one your account is not allowed to read, which is information rather than an error. Use the two dropdowns to filter by source and by data class, and the search box to filter by name.

The data class matters when you write up. `MEASURED` came off an instrument or is a direct record of something that happened. `DERIVED` was computed from something else. Nothing in this workspace is `MODELLED` or `ESTIMATED` today, and that is deliberate.

3.6 Geospatial Layers

WHAT IT IS FOR

Everything this account can see, on the ground. This is the view to open before creating a mission, and the one to open when a finding needs to be understood in place.

WHAT IS ON IT

LAYER	WHAT IT SHOWS
Frame footprints (green)	Where each geolocated archive frame was pointing. Click one for the record and a thumbnail.
Frame centres (blue dot)	The coordinate itself, which is what the pipeline tests against a mission area.
Mission areas (white dashed)	Your own missions. Click one to open it.
Findings (amber)	Every candidate zone and map layer any of your runs produced. Click for the numbers behind it.
Area of interest (red)	The rectangle you have set with the button. It is carried into the New Mission dialog.

Three basemaps are offered in the control at the top right. **Satellite (Sentinel-2)** is real satellite imagery of Kuwait and is the default. **Street map, dark** and **Street map** are OpenStreetMap, which carries the place names and roads that imagery alone does not. Any layer can be switched off with its tick box.

HOW TO USE IT

- **Fit to frames** zooms to the ground KuwaitSat-1 has actually covered.
- **Fit to my missions** zooms to everything you have defined.
- **Use this view as an area of interest** takes the rectangle currently on screen, clips it to Kuwait, and remembers it. Open New Mission next and the map starts there.

- The read-out in the bottom left gives the centre coordinate, the zoom and the width of the view in kilometres. The address bar follows the map, so you can copy the link out of it and send a colleague the exact view you are looking at.

3.7 Research Console

WHAT IT IS FOR

Running the pipeline and watching it decide. This is where the work happens.

HOW TO USE IT

- 1 Choose a mission from the dropdown. Its recorded objective appears below, read only: the objective belongs to the mission, so that a run can always be traced back to the exact question that was asked. To ask a different question, create a different mission.
- 2 Press **Run Analysis**. The line under the dropdown narrates what the Orchestrator is doing. The **Agent Execution Trace** at the bottom fills with steps.
- 3 Read the trace. Each row is one agent, the tool it called, whether it was allowed, and when. Every one of those rows was written to the database before it appeared on your screen, and the screen is showing you what the database returned. It is the audit trail, not a script.
- 4 The run stops at the **Human checkpoint** panel. See section 5.

AGENT TOOL PERMISSIONS

The panel on the right is not a list of intentions. Every run attempts the forbidden write to the payload archive on purpose and records what the database said back, as a refused step in the trail. You can see that row in Provenance / Audit after any run. If a run ever reports that the write was *not* refused, stop and tell the team: it would mean a permission had been widened.

3.8 Reports & Exports

WHAT IT IS FOR

The outputs that leave the workspace.

CONTROL	WHAT IT DOES
The report table	Every report approved on your account, newest first. Press Open to read one.
Download .md	Saves the open report as a Markdown file. It opens in any text editor, Word, or a Markdown viewer.
Print	Prints just the report, on white paper with black ink, without the workspace around it. Choose "Save as PDF" in the print dialog to get a PDF.
Export findings as JSON	Saves every finding on your account as GeoJSON. It opens in QGIS, ArcGIS or any text editor, with the geometry attached.
Export frame record as CSV	The acquisition metadata for the archive. Opens in Excel.

A report exists only because a person pressed a button. It carries the account that approved it and the moment it was approved. The ten sections it contains are listed on the view and are the same ten every time, so two reports from this platform can be read side by side.

3.9 Provenance / Audit

WHAT IT IS FOR

The receipt. Every automated run on your account leaves a row here, including the runs that were refused and the ones you abandoned. Nothing is deleted.

PANEL	WHAT IT TELLS YOU
Runs	Every run: which mission, its status, when it started and finished, and how many tool calls it made.
Agent steps	Every step of every run. Use the dropdown to narrow to one run. The Arguments column carries the refusal reason where there is one.
Steps refused	A count. A healthy number here is not zero: each refusal is evidence that a boundary held.
Access test	Four requests your account is not supposed to be able to make, sent for real against the live database, with the answers printed verbatim.

THE ACCESS TEST

Press **Run the access test** and the workspace attempts to modify the payload archive, insert a frame into it, write directly to the audit trail without going through the checked function, and change a mission's state from the browser. All four should come back refused, with the database's own words. The test also reads the

researcher profile table and reports how many rows came back: one, your own, however many researchers are enrolled. That single row is row level security working.

Nothing in this test is simulated and nothing is caught before it leaves. Run it in front of anyone who asks how the platform is secured.

4 A complete mission, end to end

This is the whole workflow in one pass. It takes about five minutes.

- 1 **Look at where the data is.** Open **Geospatial Layers** and press **Fit to frames**. The green rectangles are the ground KuwaitSat-1 has photographed. Pick one you want to study.
- 2 **Frame the ground.** Pan and zoom so the map covers that footprint with some margin, then press **Use this view as an area of interest**.
- 3 **Create the mission.** Press **New Mission**. Name it. Write the objective as a question, for example: *Identify areas inside Kuwait Bay where increasing vegetation could improve environmental conditions, and state the data, assumptions, uncertainty and limitations*. The map already shows the area you set. Press **Create mission**.
- 4 **Run it.** On the mission that just opened, press **Run the pipeline on this mission**. You land in the Research Console and the run starts.
- 5 **Watch the trace.** The Satellite Data Agent reports how many frames were released, how many are geolocated and how many fall inside your area, then attempts the forbidden write and records the refusal. The Environmental Analysis Agent measures each frame inside the area. The Recommendation Agent ranks the bare ground.
- 6 **Decide.** The run stops. Read the candidate zones in the Human checkpoint panel: each carries its greenness index, its luminance, how much of the ground around it already carries vegetation, and roughly how many square kilometres it covers.
- 7 **Approve, re-rank or abandon.** See section 5.
- 8 **After approval**, the Impact Prediction, Visualization and Reporting agents run and the mission moves to `REVIEW`. A draft report now exists.
- 9 **Read the draft.** Open **Missions**, click the mission, and read it under Findings. If it is not right, that is a signal to change the area or the question and run a new mission, not to edit the draft.
- 10 **Sign it.** Press **Generate the report**. Your account is recorded as the approver and the mission closes. The report is in Reports & Exports, where it can be downloaded, printed or saved as a PDF.

5 The six agents and the human checkpoint

The Mission Orchestrator is not one of the six. It is the control flow: at every stage it reads what the previous stage actually returned and chooses the next permitted action from it. Three of its branches end the run early, and those are the interesting ones.

AGENT	WHAT IT DOES, AND WHAT IT IS ALLOWED TO DO
Satellite Data Agent	Reads the payload archive and reports what evidence exists inside the mission area. Allowed to search and retrieve. Not allowed to modify anything, and it proves that on every run.
Environmental Analysis Agent	Measures each frame inside the area. It divides the frame into an 8 by 8 grid and computes the mean colour of every tile, classifying each as water, vegetation or bare ground against two written thresholds.
Recommendation Agent	Ranks the bare tiles and returns the best five. It proposes. It does not decide.
Impact Prediction Agent	Runs only after you approve. It reports a range or states plainly that no estimate can be made. See below.
Visualization Agent	Composes the map layer: the mission area, the footprint of every frame used and the candidate zones.
Reporting Agent	Assembles the ten section draft from the measurements. It writes a draft. Only a researcher turns a draft into a report.

WHAT HAPPENS AT THE CHECKPOINT

YOUR CHOICE	WHAT THE PLATFORM DOES
Approve the candidate set	The last three agents run and the mission moves to REVIEW with a draft report attached. Your approval is recorded in the trail.
Reject and re-rank	The candidate set is <i>rebuilt</i> from the same measurements under a different criterion: driest ground first rather than bare ground beside existing greenery. The first set is not deleted. Both stay in the findings and the trail shows both rankings, so a reader can see what was considered and what was chosen.
Abandon this run	The run closes as stalled. Every step that did happen stays in the audit trail. Nothing is deleted, ever.

Why the pipeline stops here and not somewhere else. Everything before the checkpoint is measurement. Everything after it is a claim about what should be done. The platform will measure on its own; it will not make a recommendation in your name without you. That is why the run is deliberately left open while it waits, and why the report carries the account of the person who approved it.

6 When the pipeline stops early, and why that is correct

The objective reads as an instruction

The Orchestrator screens the research objective before it calls any tool. An objective that contains an instruction to disregard the rules, a reference to a credential, SQL that modifies data, an instruction to send data to an external address, or markup that would execute if it were rendered, is refused. No frame is read, no finding is produced, the mission is marked with an injection flag and the refusal is written to the trail.

Rewrite the objective as a question about the data and create a new mission.

The screen is a detection heuristic, not the defence. What actually keeps a hostile objective harmless is that nothing downstream of it can do anything dangerous: the archive carries no write permission for any signed-in account, there is no outbound network call in the pipeline at all, and the database decides what any caller may read. The screen exists so that the attempt is *visible*, not so that it is stopped. It was already stopped.

No archive frame falls inside the mission area

The run stops at the first step and records that there was nothing to measure. It does not substitute a frame from somewhere else and label it with your area. Open Geospatial Layers, see where the frames actually are, and create a mission over one of them.

No impact estimate is produced

This is the most important one to understand, because it will happen on every mission you run against the archive as it stands today, and it is the correct result.

Estimating what a change would do requires observing the same ground on at least two different dates. The Impact Prediction Agent counts the repeat visits that actually exist in the frames it used. On the current archive there are none: every frame is a single pass over different ground. So the agent states the candidate area, the surface class of each tile and the date each measurement was taken, and then says plainly that no rate, trend or projected effect can be given.

That is not a gap in the finding. It *is* the finding, and it is the sentence that makes the rest of the report trustworthy. A platform that produced a percentage here would be inventing one.

7 What the measurements can and cannot support

Every report the platform produces carries these limitations in its own words. They are repeated here so you can quote them before you have run anything.

THE GREENNESS INDEX IS A PROXY, NOT A VEGETATION MEASUREMENT

KuwaitSat-1 carries three visible bands and no near infrared band. NDVI, the standard vegetation index, needs near infrared and therefore cannot be computed from these frames. The platform uses Excess Green

on the visible bands instead, which is a published visible band proxy. The frames are JPEG, raw decoded, with no radiometric or atmospheric correction and no illumination normalisation between passes. Values are comparable *within* one frame and not *between* frames. Do not read a vegetation percentage off them.

FOOTPRINTS ARE AN INDICATION OF EXTENT, NOT A SURVEY BOUNDARY

A footprint is computed from the frame centre, the 39 metre ground sample distance and the pixel dimensions. The payload record carries no attitude, so north is assumed to be up and no rotation is applied, and the ground is treated as flat at the frame centre. Both are approximations and both are stated on every footprint the platform draws.

A FRAME COUNTS AS INSIDE THE AREA WHEN ITS CENTRE IS INSIDE

Frames that overlap the edge of your mission area are not included. Draw the area a little wider than the ground you care about.

THE GEOLOCATIONS ARE OPERATOR JUDGEMENT

They were matched by eye against reference imagery. The confidence on each frame is that judgement, not a measurement. Cite the confidence when you cite the coordinate.

THERE IS NO GROUND TRUTH

No tile in any run has been checked against an observation on the ground. Every classification in this platform is an inference from colour.

8 If something goes wrong

WHAT YOU SEE

WHAT IT MEANS AND WHAT TO DO

This workspace is restricted	No session was recognised. Reload once, then sign in again from the mission hub.
The archive refused this read	Your account reached the database and was declined. Usually it means enrolment has not completed. Sign out, sign in again from the mission hub, and tell the team if it persists.
The database refused this insert	A field failed a check. The message names the reason. If it mentions the area, move the map back over Kuwait.
This mission is already running	A run is open on it. Open the mission and use "Open the checkpoint in the console", then approve or abandon it.
This mission has an approved report	The mission is closed on purpose. Evidence behind a signed conclusion must not be able to change after it is signed. Create a new mission.
Limit reached: N agent runs per hour / per day	A rate limit, applied per mission and per account. Wait, or use the time on a different mission.
The prototype is not accepting new missions right now	Runs have been paused centrally. Ask the team.
A report needs at least 50 characters	You pressed Generate the report with no draft behind it. Run the pipeline and approve the candidate set first.
A table is empty and says so	That is the database answering with nothing, which is a real answer. An empty mission list means you have not created one; it never means somebody else's work is hidden from view by accident.
The map is blank	The basemap tiles come from the internet. Check the connection, then try a different basemap in the control at the top right.

9 Words used in this workspace

TERM	MEANING HERE
Mission	A research question plus a polygon over Kuwait. Everything else hangs off one.
Run	One pass of the pipeline over one mission. A mission can have several.
Step	One agent doing one thing, recorded whether it succeeded, was refused, or failed.
Finding	Something a run produced: a candidate site, a measurement, a map layer or a narrative.
Frame	One picture KuwaitSat-1 returned, with its acquisition record.
Footprint	The ground a frame covers, computed rather than recorded.
GSD	Ground sample distance. 39 metres per pixel on this payload: one pixel is 39 metres on the ground.
ExG	Excess Green. A visible band greenness proxy, used because there is no near infrared band.
Area of interest	The polygon on a mission. Always inside Kuwait, always four corners, always checked by the database.
Human checkpoint	The point where the pipeline stops and a person decides.
Row level security	The database rule that scopes every table to your account. It is why your work is private.
Injection flag	A marker on a mission whose objective was screened as an instruction rather than a question.

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